



DEALER BEST PRACTICE SERIES

Backlog Management

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

This Best Practice deals with the creation, management, and reporting of maintenance and repair Backlogs. Implementing this Best Practice will improve machine availability and reduce operating costs by the reduction of unplanned, costly repairs. It will also provide a method to identify, manage and schedule necessary repairs before failure.

This Best Practice addresses the items in **Site Assessment Section 2.8**.

Backlog is generally understood to be the work that has not been completed by the nominated "required-by-date". Backlogs are also managed as pending workload for the repair centers. Our approach to Backlog Management is to view it as a powerful tool to proactively prevent failures. Backlog management is one of the keys to a good equipment maintenance system. If the effort that goes into condition monitoring does not result in a high percentage of planned and scheduled repairs, the corresponding results of reliability, costs and availability will be adversely effected. The planning and scheduling routines will also be disrupted.

2.0 Best Practice Description

Backlog Management

Backlog management is one of the keys to a good equipment maintenance system. If the effort that goes into condition monitoring does not result in a high percentage of planned and scheduled repairs, the corresponding results of reliability, costs and availability will be adversely effected. The planning and scheduling routines will also be disrupted.

Backlog is generally understood to be the work that has not been completed by the nominated "required-by-date". Backlogs are also managed as pending workload for the repair centers. Our approach to Backlog Management is to view it as a powerful tool to proactively prevent failures.

It is commonly understood that there are certain types of failures that will occur, with no signs or measurable warning. Examples are those related to the electronic or electrical control systems. Fortunately our experience shows that most of the failures that mobile equipment experience that cause significant downtime are detectable well in advance of failure, allowing the organization to plan the appropriate corrective actions.

There are two areas that we would like to stress: the early detection of possible failures and their appropriate corrective planning actions. Early detection of potential problems increases the time window to plan the necessary repairs.

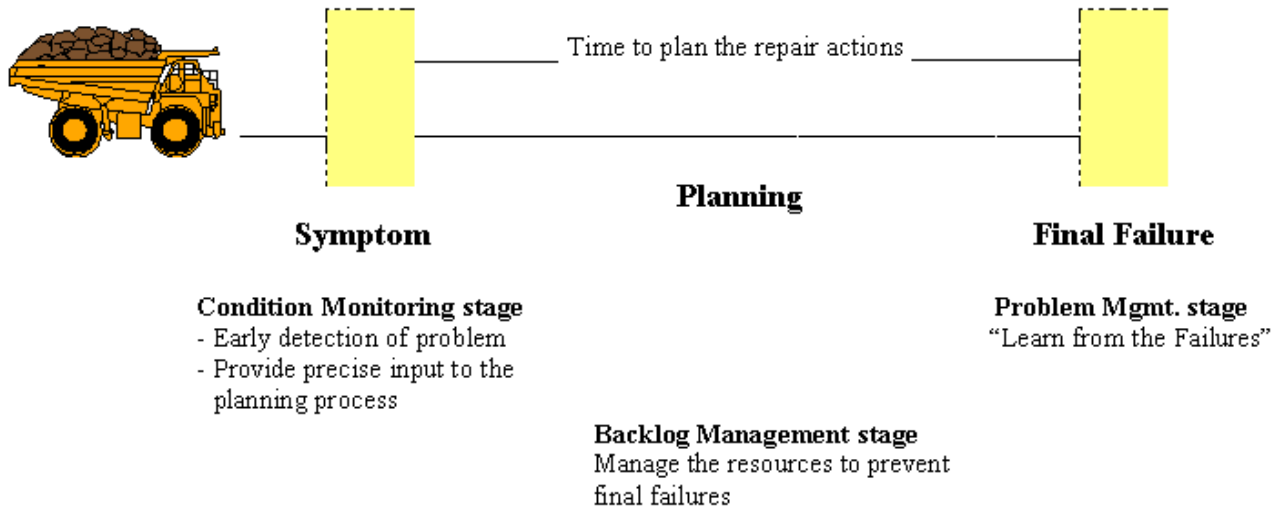
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We would like to use this diagram above to illustrate our approach to Backlog Management. Backlog Management is the process occurring between the detection of a symptom of a potential or hidden failure and the occurrence of the actual failure itself.

Condition Monitoring and Backlog Management are so critical to achieving the goal of reliability, that any maintenance organization must be able to assess and improve the process. Establishing a logical approach for this planning stage is probably the first step.

We would like to contribute to this effort by presenting a process that was developed in conjunction with our mining dealers.

2.1 Backlog Management Process

2.1.1 Generating the Repair Request.

The effectiveness of Backlog Management begins with the information that is provided by the maintenance inspector, the operator, the Fleet Analyst, or whomever inspected the equipment at the time the repair needs are determined.

As a first step in defining the process we recommend adopting a unique form to communicate the repair need, designing the form in accordance with the desired end result. This form will be used primarily by the inspector, field and shop mechanics, supervisors, and should be available to anyone that needs to enter a repair request. The results of the operators' input need to be filtered, normally done by an inspector, to transform their request into a formal repair request.

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As soon as a repair request is generated, it becomes a Backlog (BL) in our process, triggering the planning actions. It is also added to the database and is introduced into the general evaluation process.

	Description	Purpose (Desire End Result)
Equipment ID	Mine internal ID commonly used	Tracking and statistical evaluations
Date	Date when Backlog was generated	Evaluate the response time of the system (age of Backlogs)
Reference Number	Number assigned in Planning	Tracking
Work Order Number	Optional. Used if Work Order is opened as soon as the BL is generated.	Enters the BL to the Maintenance Support Software
Type of Inspection	Identifies the generation source (Condition Monitoring routines + operational areas, such as Field, PM, Shop, others)	Evaluate the contribution of the different organization areas
Symptom	State the problem	Identify the warning signs that were detected that indicated the need for a specific repair
Repair Action	Describe the repair action needed	Identify a precise repair need eliminating further inspections
System	Identifies the major systems of the equipment (Use of SMCS codes recommended)	Statistical Analysis. - Actual BLs per system - BLs per system generated
Priority	Identifies the urgency	Planning. Treat the BLs accordingly to their urgency
Estimate of Repair Times	Identifies the Labor hours and Downtime estimated for the repair	Estimate criticality of Backlog and future shop work load
Parts request	Identifies the parts needed for the repair	Parts must be identified by the generator (man on the machine)
Tooling, Special Equipment, Consumables Requested by	Identifies special needs for tooling, equipment, consumables, information, etc.	Efficient Planning.
Supervisor approval	Identifies the generator. Especially important when operator is involved.	Facilitate possible verification. Recognition & motivation.
Keypunched OK	Supervisor is responsible for the quality of the BL generated	Efficient Planning.
Parts estimated arrival date	Identifies when the BL is entered into the database	Process control
Parts arrival date	For Parts department use.	Planning & control.
	Identifies the actual parts arrival date	Triggers the next planning stage, schedule the repairs.

Accurate and complete information permits the planning department to perform their functions effectively and efficiently, avoiding further effort duplication due to reevaluation of repairs needed.

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The following Backlog Record Form is proposed for gathering the list of information required. This form is currently in use by various operations that operates Caterpillar ☐ and other brands equipment.

BACKLOG RECORD FORM							
Equip. No: _____		Date _____		Reference No. _____		W. Order No. _____	
Type of Inspection: Operator ☐ Field ☐ Preventive Maint. ☐ Other ☐							
Symptom : _____							
Action : _____							
System : _____							
1000 Engine	3101 Torque Conv.	4300 Steering	7000 Basic Mach.	9500 Various			
1400 Electrical	4050 Final Drive	5050 Hydraulic	7200 Suspensions	9600 Factory Modif.			
3000 Transmission	4250 Brakes	5500 Tires	7320 A/C				
Priority : Urgent ☐ Next PM ☐ When Possible ☐ Monitor ☐							
Estimate of Repair Times							
Machine Hours Hr.		Man Hours Hr.					
Parts Request							
Item	Quantity	Part Number	Description	Page	Item	Observations	
1							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
						Order Number	
Tooling, Consumables and Support Equipment							
Lift Truck	Liquid Nitrogen						
Crane	Jack						
Tire Equipment	Washing						
Requested by: _____				Estimated Arrival Date _____			
Supervisor Approval _____				Parts Arrival Date _____			
Keypunched OK _____							

NOTE: The "Type Of Inspection" field can be modified to accommodate the condition monitoring routines that are implemented and the organization areas that may contribute with the Backlog generation.

Now that we have our entry point defined, we can go into the second stage, the planning process.

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2.1.2 Backlog Planning Process Stage

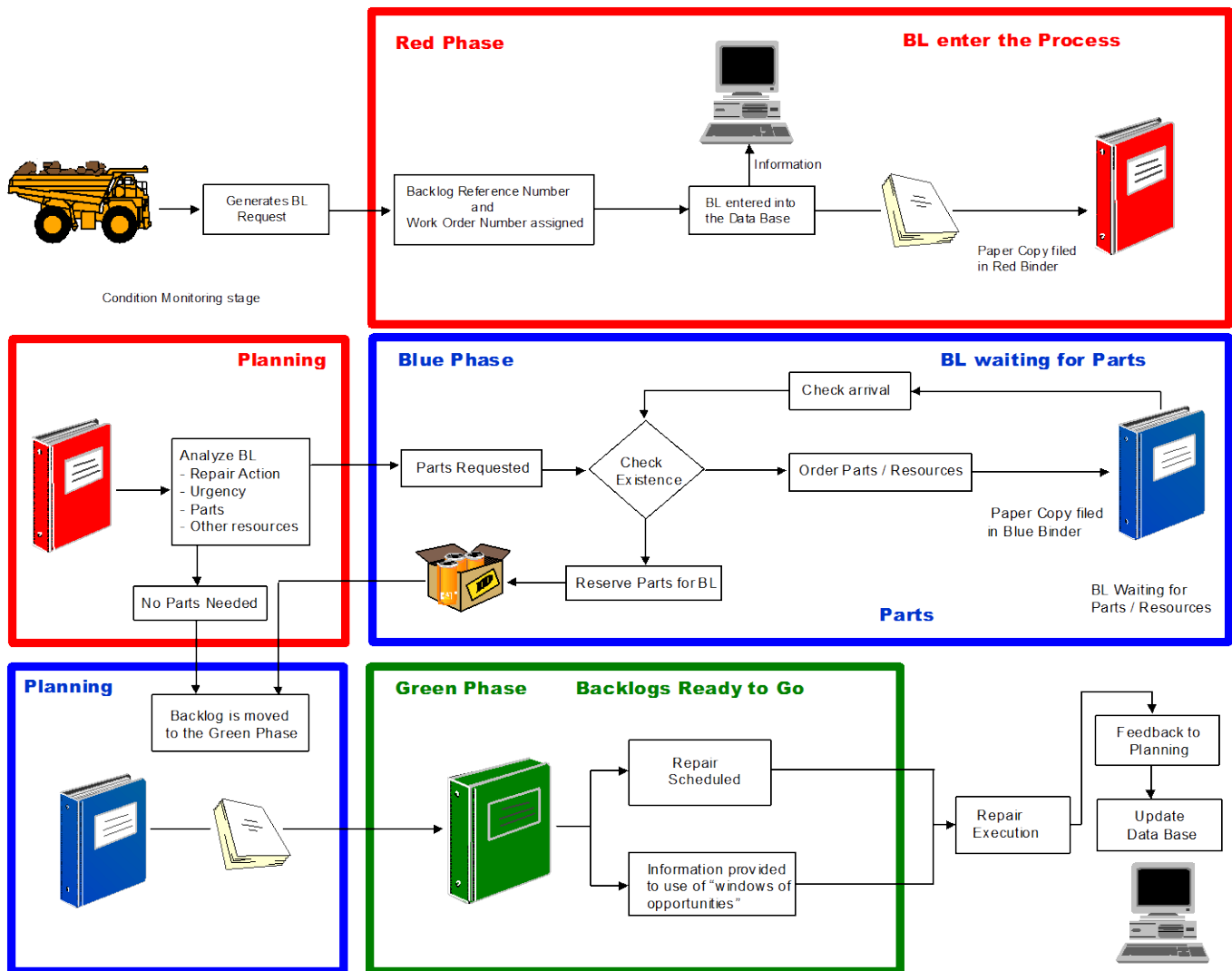
As soon as the Backlog Form is completed it enters the planning area. There are three major phases in the Backlog Management process:

- Backlog received and entered to the database. Waiting to be processed.
- Backlog planned and waiting for resources, normally parts.
- Backlog ready to be executed. Scheduled, waiting to be executed.

To facilitate the identification of these phases and the handling of the quantity of Backlogs entering the system, we assigned colors to these stages. Red, Blue and Green are the colors selected respectively.

Let's follow a Backlog through the entire process:

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The starting point of the planning process begins in the "Red stage" where the Backlog request is entered into the database. At this point the request could stay "waiting to be planned" or could pass to the next stage if the planning actions are taken immediately. As soon as the planning actions are taken, analyzed, the actual repair required is established, and the resources needed (parts and others) are ordered, the Backlog is passed to the next stage, the "Blue stage". All Backlogs in this blue stage are waiting for some resources, most frequently parts. The next stage, the "Green stage", contains all Backlogs that are "ready to go", with all the necessary resources allocated. The arrival or physical availability of parts and other special resources is the condition to move the Backlog to this last stage. At this point the Backlog is ready to be scheduled for execution or to be performed in the earliest window of opportunity.

The use of distinctive areas of functionality and responsibility helps with possible improvements determined by the continuous evaluation of your maintenance operation.

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2.1.3 Evaluating the Process

As we stated at the beginning of this section, the ultimate goal is to achieve the targets for percentage of scheduled work and equipment reliability. Keeping this final goal in mind the Backlog Management process can be evaluated, for each of the defined functional areas and assess the contribution of them in the achievement of the desired end results.

Backlog Management Performance Indicators

Indicator	Description	Purpose
Quantity of Backlogs	The number of Backlogs in the process. Repair need identified, not executed.	Evaluates the work load and risk for failures
Quantity of Backlogs generated	Number of Backlogs entered into the system	Evaluates the condition monitoring ability to detect potential problems.
Estimated Time to Repair	Sum of all repair estimated time determined at the time the Backlog was generated	Evaluates the severity of the Backlog work load and potential loss of availability
Percentage of Backlogs over 30 days	Measured from the date the backlog was generated	Evaluates the quality of the response of the system to react proactively to eliminate potential problems.
Quantity of Backlogs in the different process phases (Red-Blue-Green)	The number of Backlogs identified that are waiting for planning, waiting for parts and waiting for being executed	Identifies the weak area, where the process is being delayed.
Backlogs per type of Inspection	Backlogs generated by the operators, inspectors, PM & others.	Identifies the areas that are contributing in the failure detection.
Backlogs per system	Backlogs sorted by machine systems.	Permit the comparison between the current and the potential problems.
Backlogs generated per system	Backlogs generated sorted by machine systems.	Evaluates the ability of the condition monitoring routines to detect the critical problems.

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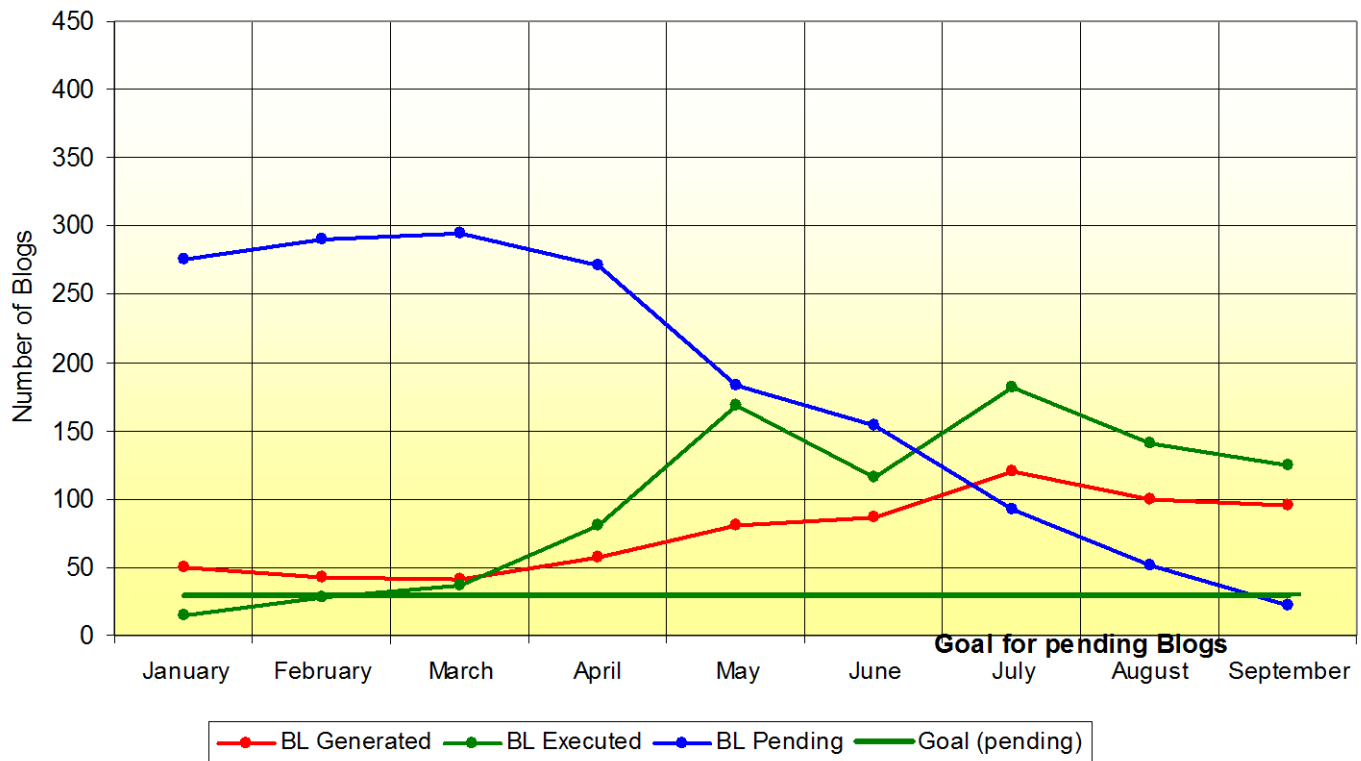
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We recommend starting by visualizing the balance between Backlog Generation, Execution and Total pending. It is also important to include the goal / target for total pending backlogs, to set up a clear management guide.

**Backlog Management - Generated / Executed / Pending
OHT Fleet**



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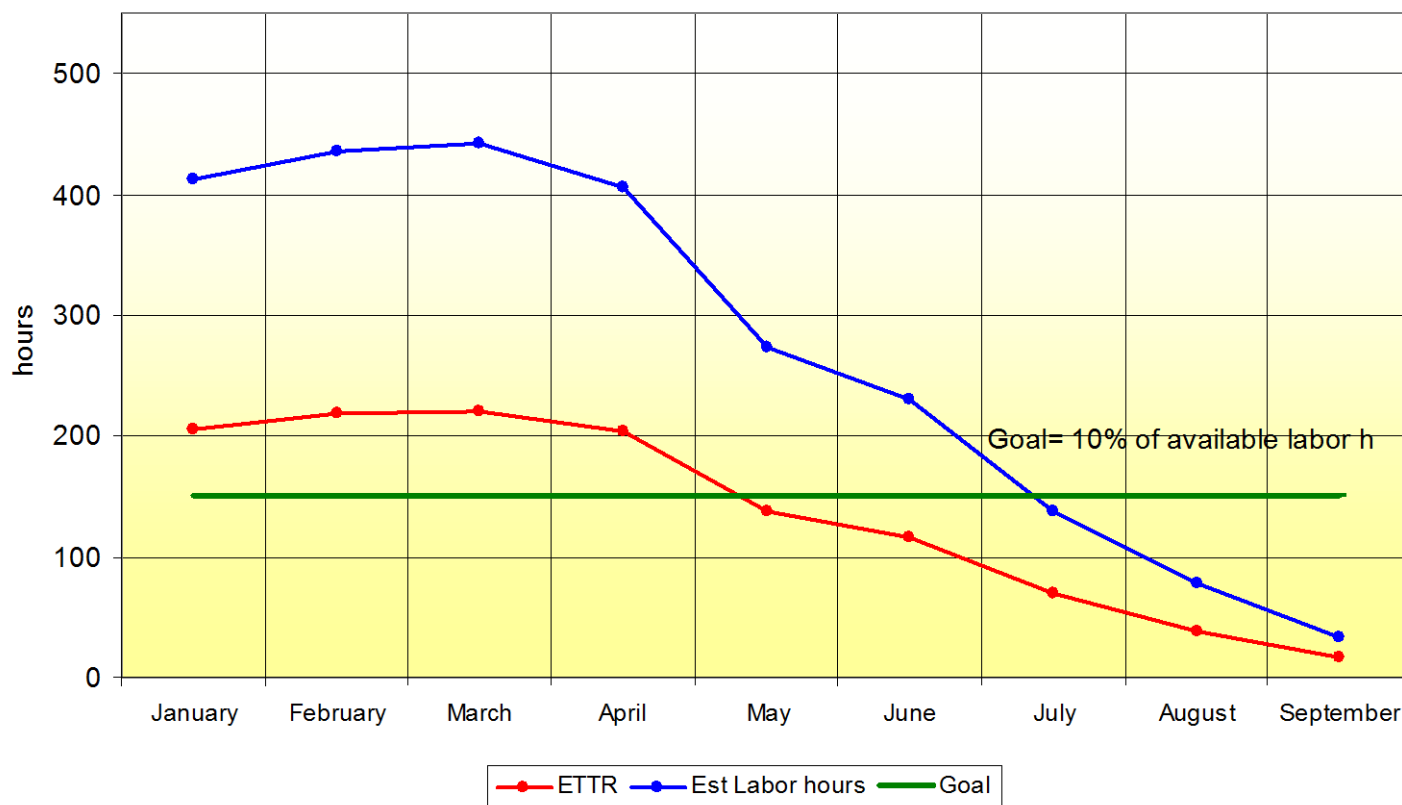
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2.1.4 Estimating Time to Repair and Labor Hours

Estimated Time to Repair in Downtime (Machine estimated hours) and labor (man power required to execute the Backlog) is a good management metric for severity of the pending Backlogs. The Goal shown in this graph is the line for 10 % of the available labor hours. It is important to maintain the hours estimated for Backlogs below 10% of the available hours on the shop floor.

Backlog Management - Estimated Time To Repair (Machine & Labor Hours)
OHT Fleet



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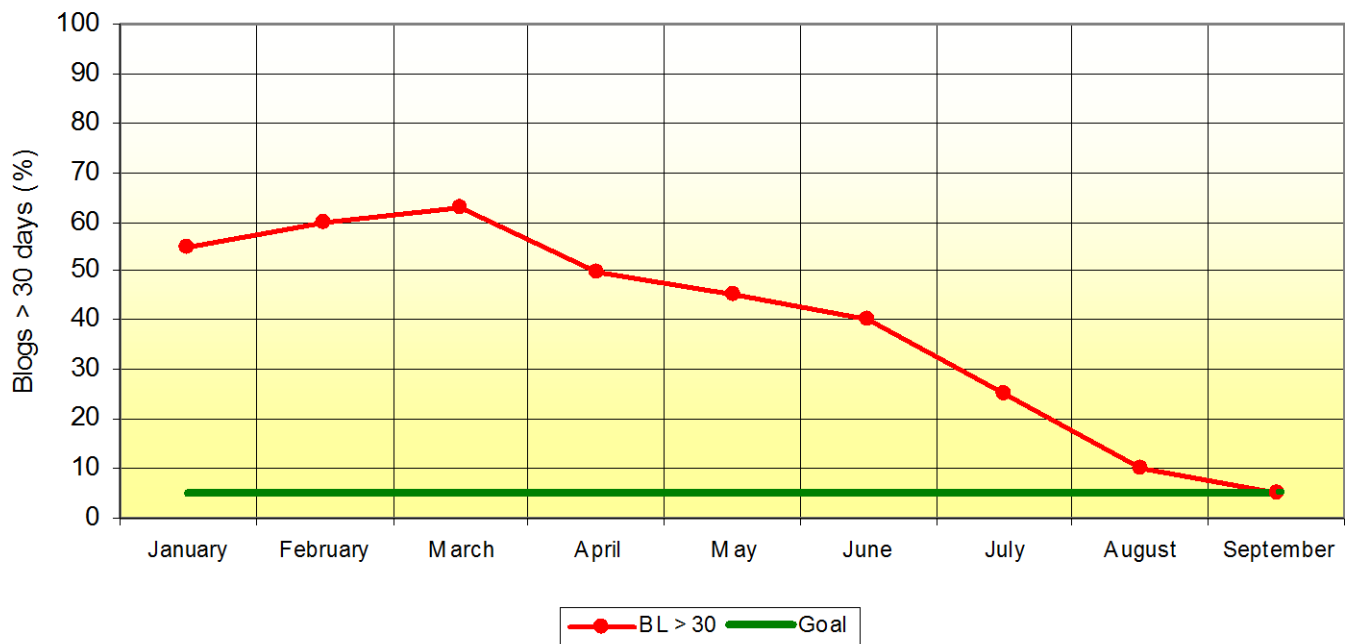
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2.1.5 Assessing Response Time

Backlogs > 30 days is a good indication for assessing the response time of your organization. 30 days is a Benchmark. We have to try eliminating the Backlog generated within 30 days. Remember that Backlogs are potential failures.

Backlog Management - Backlogs > 30 days
OHT Fleet



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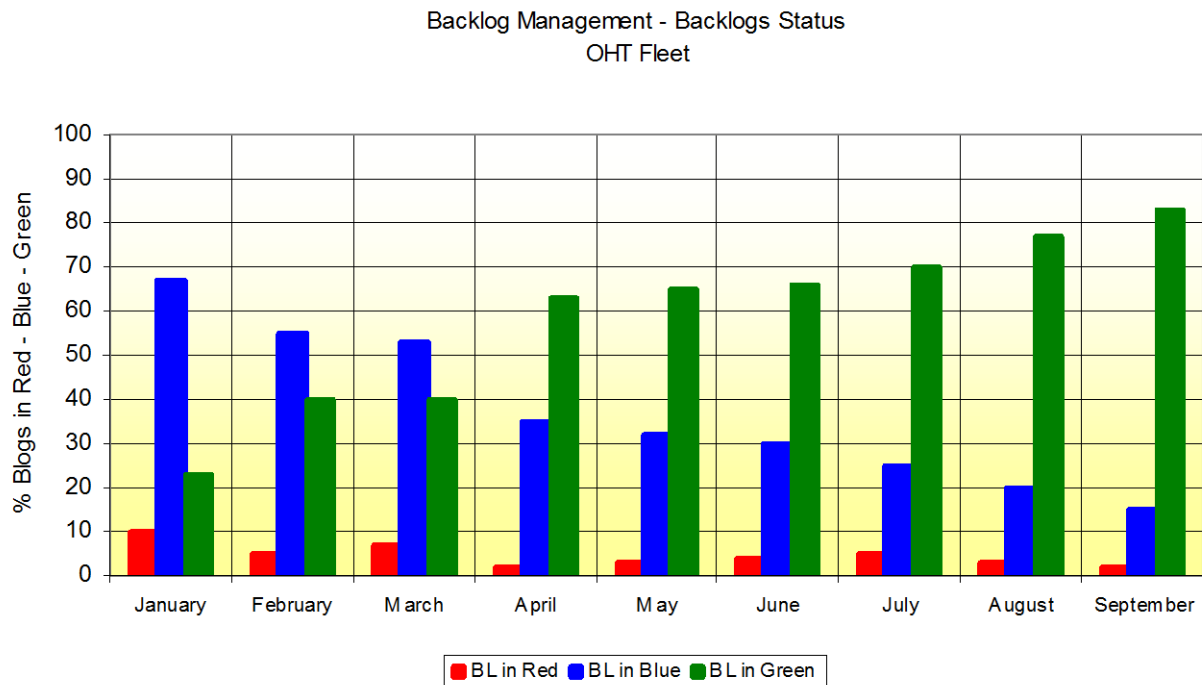
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2.1.6 Backlog Management – Backlog Status

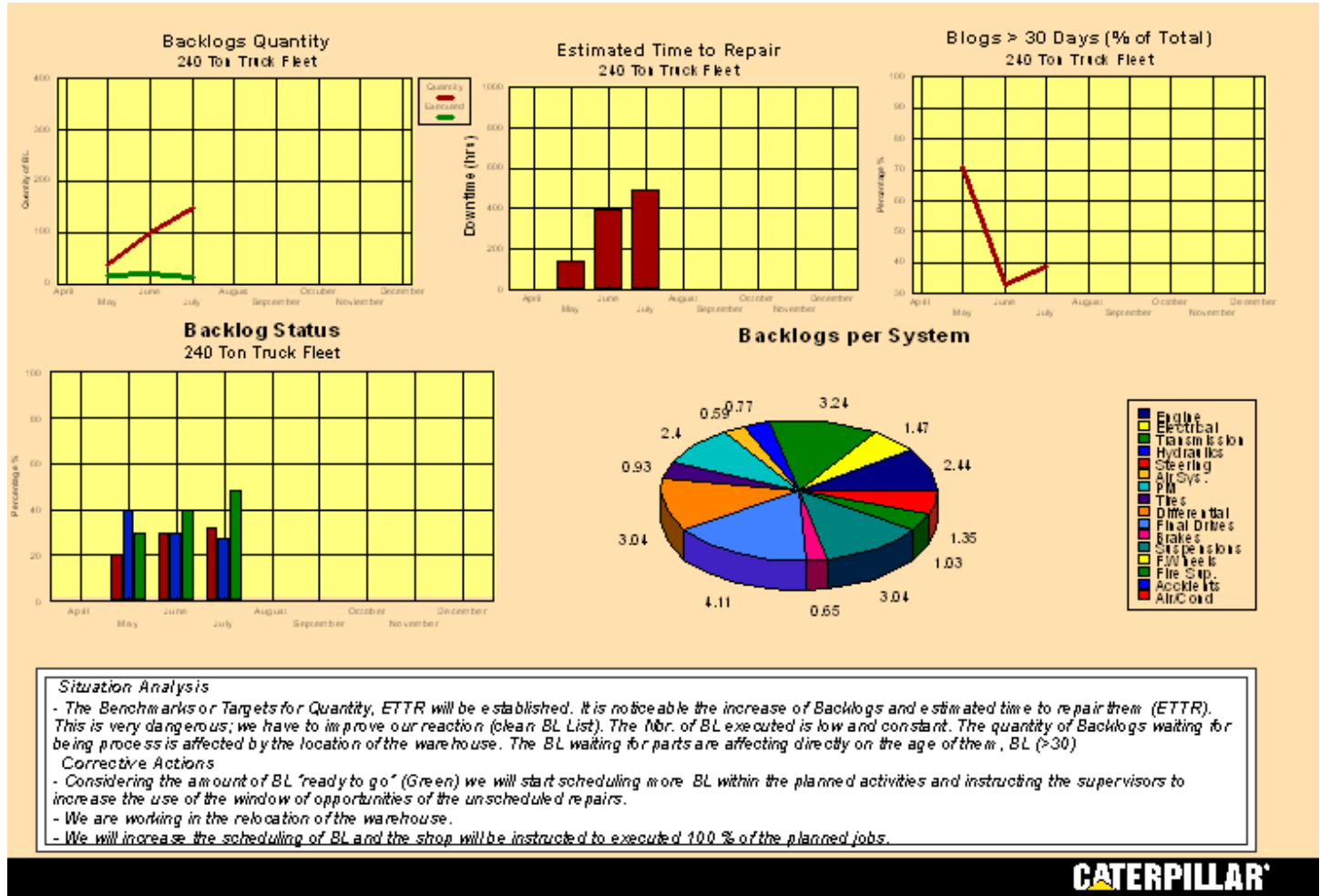
This report allows you to identify the areas of responsibility on the pending Backlogs. From all pending Backlogs, what percentage is in process in the planning area (Red), what percentage is waiting for parts / resources (Blue) and what percentage is ready to go (Green)? The managers can now concentrate on improving the appropriate area of opportunity.



Sample Backlog Management Report

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This is an example of a Backlog Management Report used in a 793B coal operation



3.0 Implementation Steps

1. Use the results on condition monitoring inspections to create Backlogs.
2. A sample Backlog form is attached.
3. Implement the 3-phase (Red, Blue, Green) method for categorizing and tracking Backlogs.
4. Use the attached evaluation process to determine the performance of your Backlog system.
5. Refer to Backlog Process Map (page 7).

4.0 Benefits

- Fewer high cost unscheduled repairs.
- Higher reliability and machine availability.
- More efficient use of shop facility and labor force.
- Lower cost per hour and cost per ton.

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5.0 Resources Required

- Backlog Record Form (attached)
- Thorough machine inspection resulting in Backlogs entered into the process.
- Three binders: (Red, Blue, Green) for tracking the Backlogs, or a computer system to duplicate this functionality.
- Individual in the Planning & Scheduling Department responsible for entering Backlogs, tracking their progress, and producing the process performance evaluation reports.

6.0 Supporting Attachments

No Supporting Attachments are included with this document.

7.0 Related Best Practices

0806-2.70-1007 -Condition Monitoring Field Inspections

8.0 Acknowledgements

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